

Airframe Technology Master

Lecture 07.

General Structures Design Principles: Mechanical Joints Part II

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Introduction

Discrete joint design criteria

- Autoalignable joint
- Non autoalignable joint
- Structural Integrity

Metallic structure joint: Failure modes

Composite structure joint:

- Solid laminate joints
- Sandwich structure joints

This lesson is intended to provide knowledge about the main type of joining methods used at aerospace sector, the parameters that should be taken into account before defining the configuration of a joint and the main criteria to take into account in the detail design of the joint.

The joints are usually classified in permanent and removable joints but there are much more parameters to be taken into account in a joint, we analyze these parameters in the next paragraph.

Definition

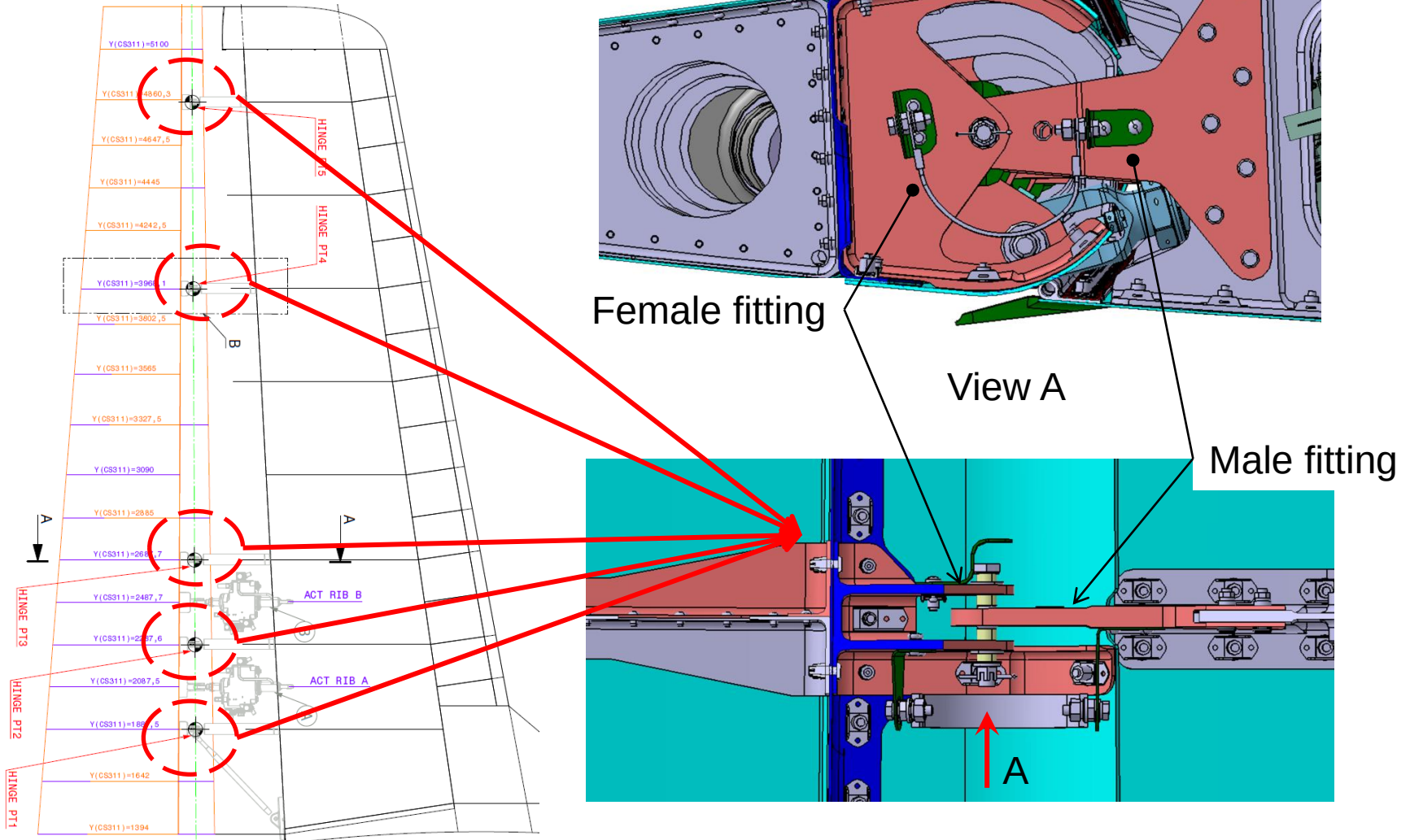
Discrete joints are used in the following situations:

- Relative rotation between the joined structures
- When one of the structures is interchangeable
- When we want to transfer the loads to the points of the structure which are designed to withstand them.
- When we want to perform an isostatic joint

The discrete joints can be divided into two main categories:

- Autoalignable joints
- Non autoalignable joints

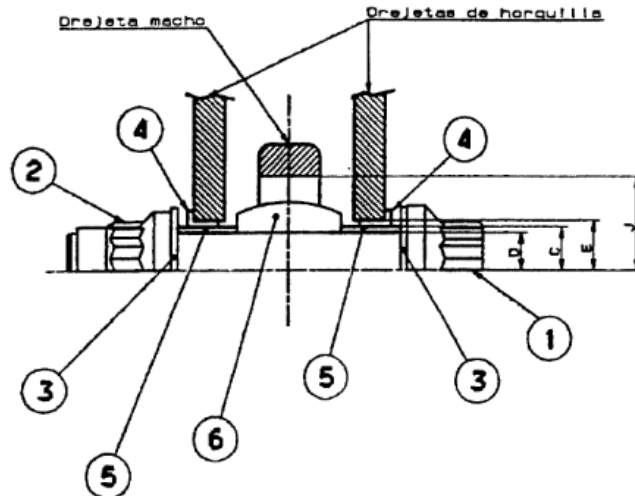
Example



Autoalignable Joint

Spherical-bearing is installed at the male lug. This joint allows misalignment between male and female fittings.

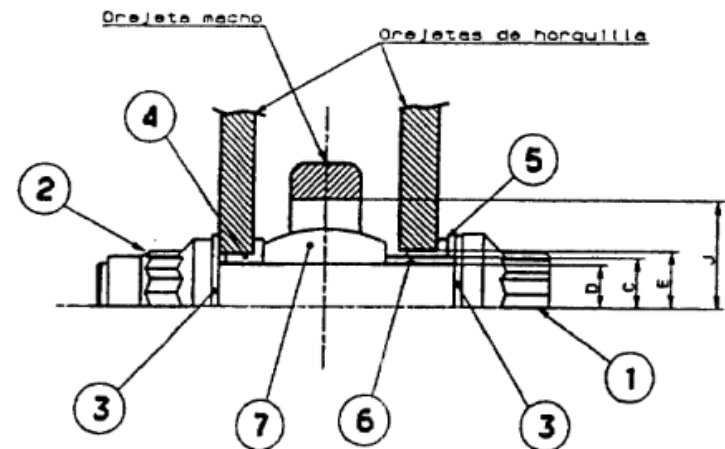
Typical joint used at movable surfaces (control surfaces, lifting surfaces, landing gear doors), allows to create a common hinge axis between several hinge fittings; one hinge fitting is clamped to carry lateral load and the other ones are not clamped



Bearing **not** clamped

Axial displacements and tolerances allowed

Bushing flange outside

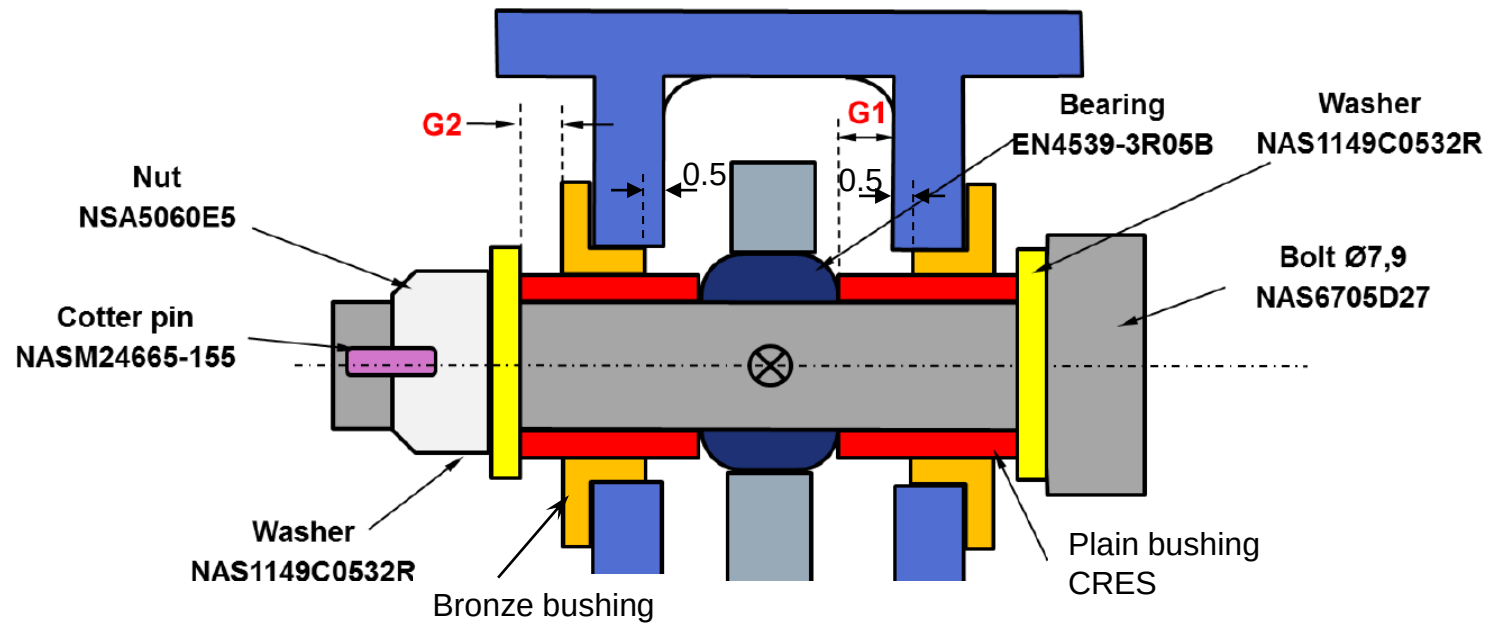


Bearing **clamped** to female lug

Axial load transferred

Bushing flange inside for bearing contact

Autoalignable Joint – Not clamped – Typical Assembly



Autoalignable Joint – Not clamped – Functional Requirements

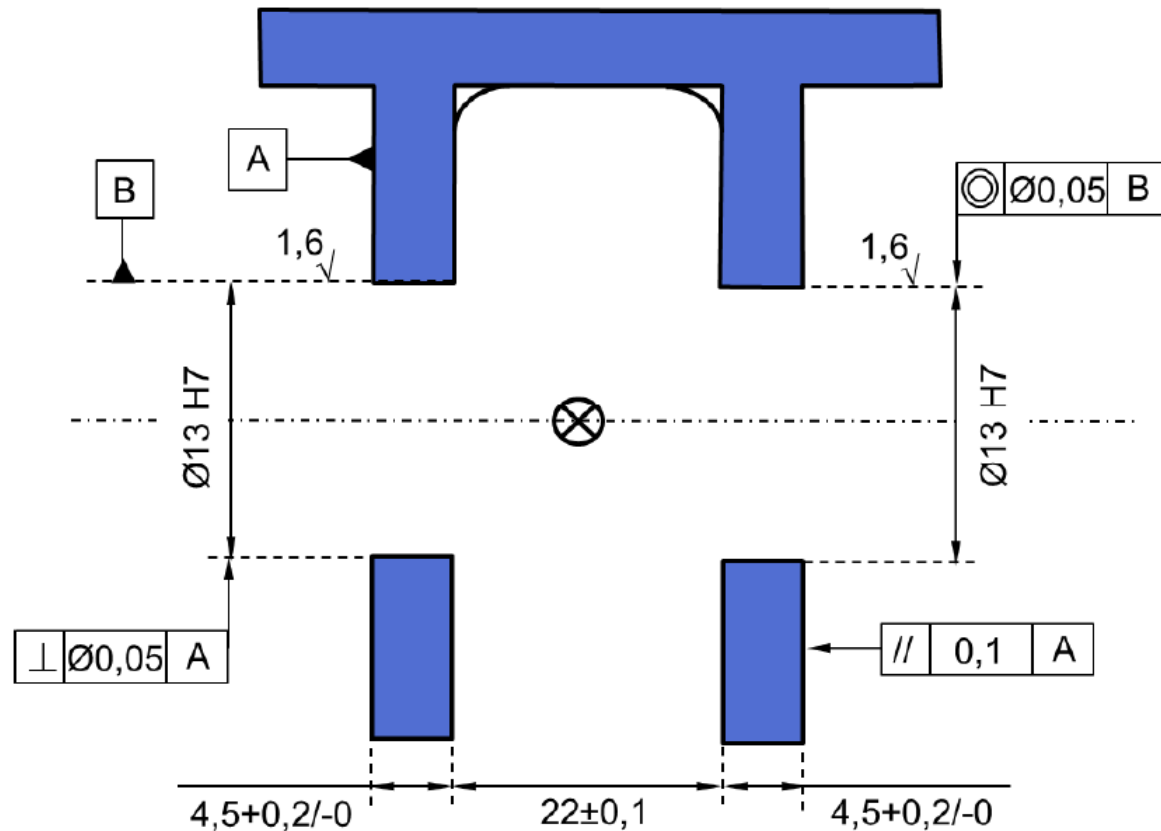
G1min > G2max + Fitting manufacturing tolerances : To avoid contact between ball bearing and female fitting.

G2 > Installation tolerances + Maximum deviation under load from clamped HF to the rest of HF's

Flanged bushing ends 0,5 mm before reach fitting surface: To provide space to sealant fillet and to avoid contact between flanged bushing and bearing avoiding bushing migration.

Ream at assembly flanged bushings: To get H7 hole tolerance, sometimes interference fit can lead to bushing deformation at assembly, and also to get concentricity tolerances between flanges holes to allow sliding-rotating fit between flanged bushing and plain bushing and between plain bushing and bolt.

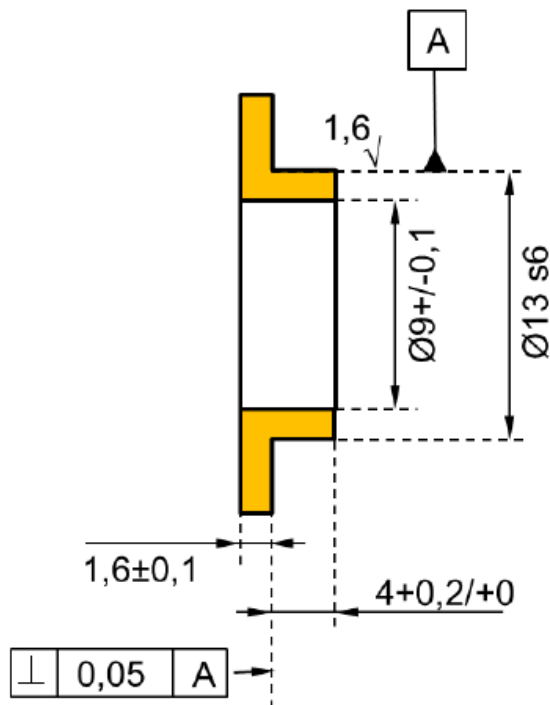
Autoalignable Joint – Not clamped – Female fitting example



Material: Aluminium 7475T7351

Finish: TSA

Autoalignable Joint – Not clamped – Bushings example

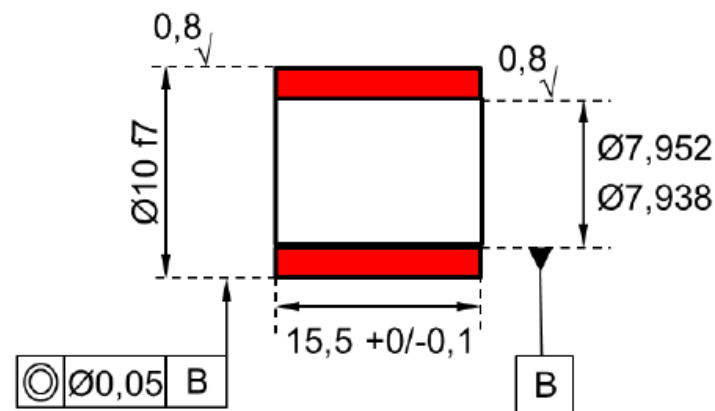


Material: Bronze C63000ANLD

Finish: Cadmium plated

Inner diameter not definitive in order to ream at assembly to H7

Outer diameter s6, H7/s6 interference assembly



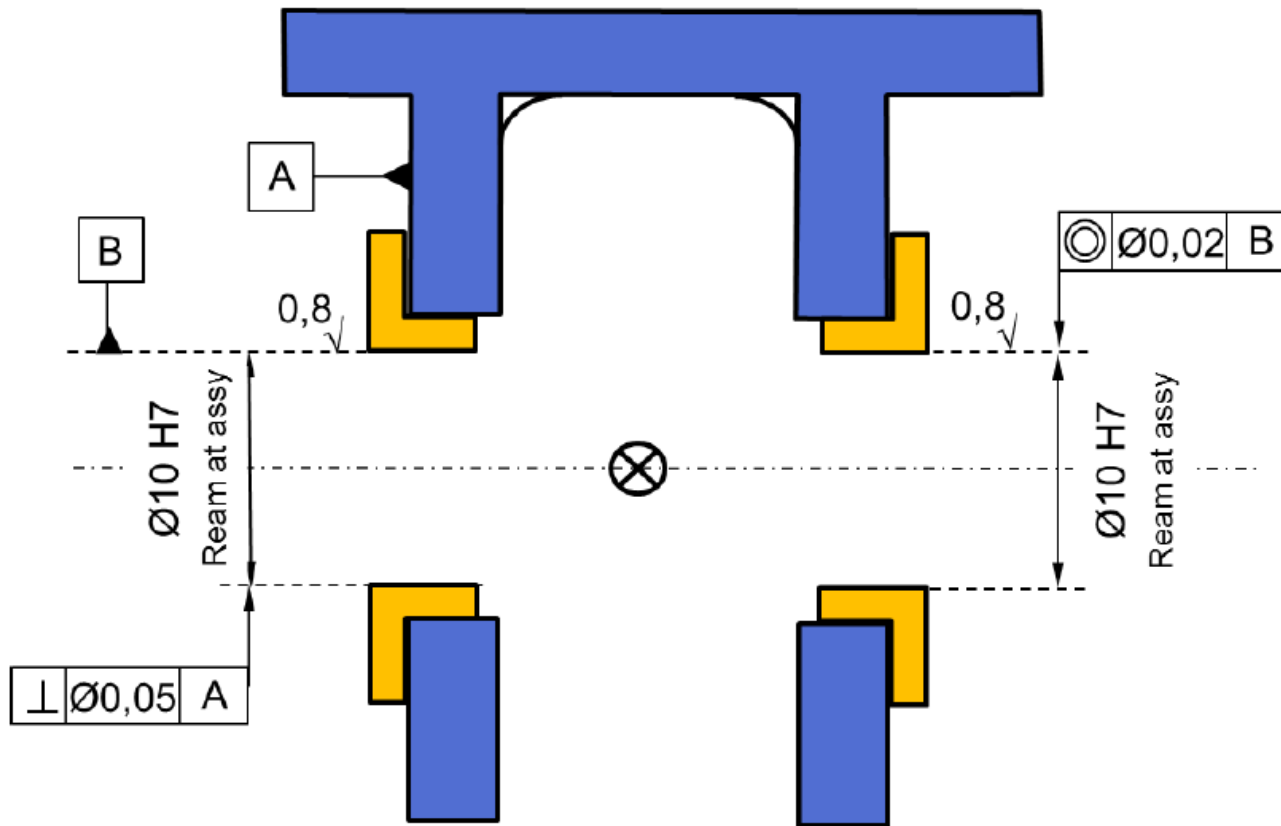
Material: PH13-8Mo-H1050

Hard chromium at outer D / Pasivated inner D

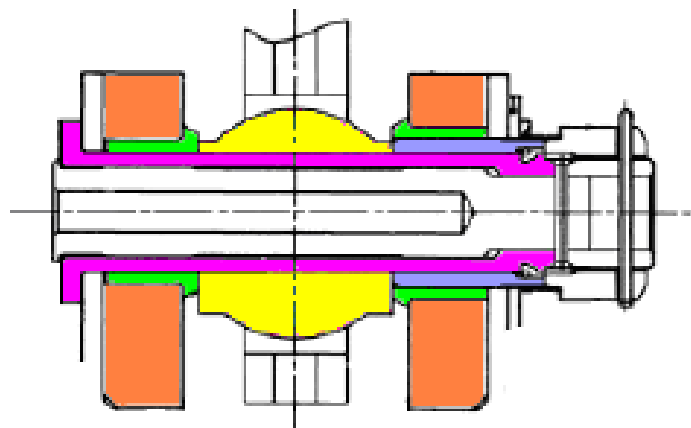
Outer diameter sliding/rotating clearance fit H7/f7

Inner diameter tolerances defined according to standard bolt

Autoalignable Joint – Not clamped – Female fitting assembly



Autoalignable Joint – Fail Safe Joint Materials & protective finish



Tungsten Carbide

	AXIS <u>Material:</u> CRES or Titanium alloy <u>Surface treatment:</u> WC-C:H coating according to AIPS 02 04 007 (see Note 1) <u>Remark:</u> Contact pressure < 500MPa Displacement: 20µm < d < 300µm (Wear mechanism: fretting wear) (example: displacement due to vibrations)
	BUSHES <u>Material:</u> CRES or Copper alloy <u>Surface treatment:</u> Zinc Nickel coating according to AIPS 02 04 006 on external shape at least
	SPACER <u>Material:</u> Copper alloy or CRES <u>Surface treatment:</u> Zinc Nickel coating according to AIPS 02 04 006 or IVD Aluminium according to AIPS 02-03-006 on external shape at least
	FITTING <u>Material:</u> Aluminium alloy <u>Surface treatment:</u> Coating should be chosen according to TN A00710003 (Note 2)
	SPHERICAL BEARING Copper or CRES standard bearing

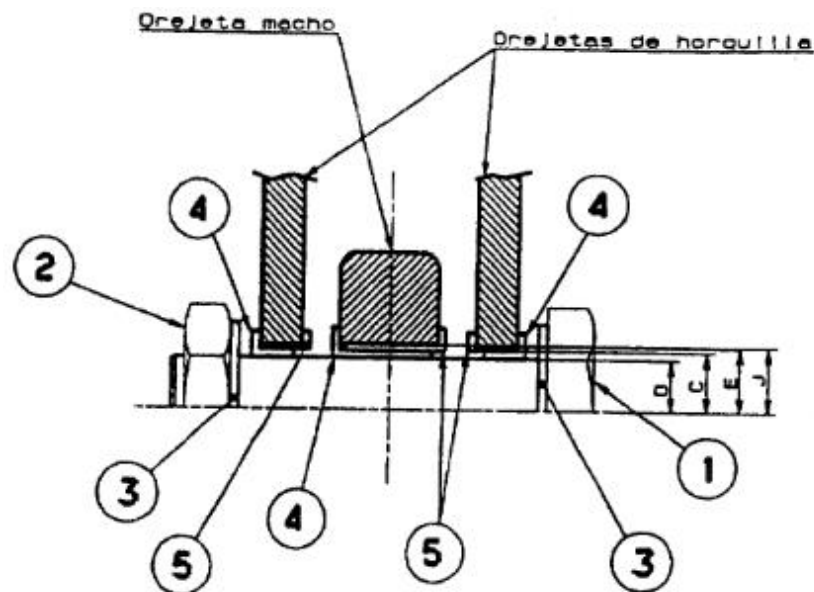
Not Autoalignable Joint

Typically used for fixed joints (HTP, VTP, wings, engine)

No misalignment allowed, high stiffness structures required

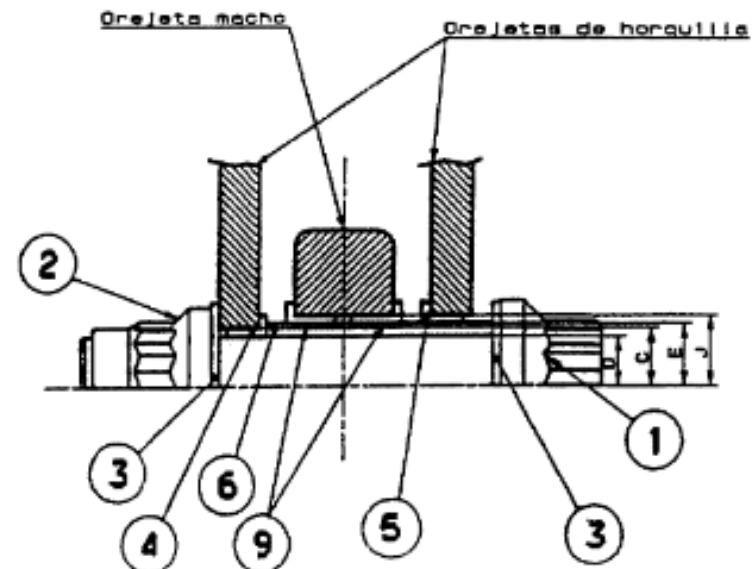
Maximum two attachment points if the joint is movable (rotation)

Flanged bushings overlapped at the male lug



Not clamped

Flanged half-bushings at the male lug



Joint clamped to female lug

Tolerances

Clamped Bushing:

Hole: H7

Bushing external Diameter: s6

Sliding/rotating bushing:

Hole: H7

Bushing external Diameter: g6 (for movable surfaces with several hinge fittings the bushing tolerance should be evaluated taking into account the fittings assembly tolerances)

Holes coaxiality tolerance can be achieved by reaming the clamped bushing once installed at the fitting, in order to do this the bushing is manufactured with an inner diameter smaller than the final hole diameter.

Materials

Male/Female do not have relative movement:

- Clamped Bushing: CRES

- Plain bushing: CRES

Male/Female have relative movement:

- Clamped bushing: Bronze to avoid jamming

- Plain bushing: CRES with Hard Chromium Surface Protection to avoid wear

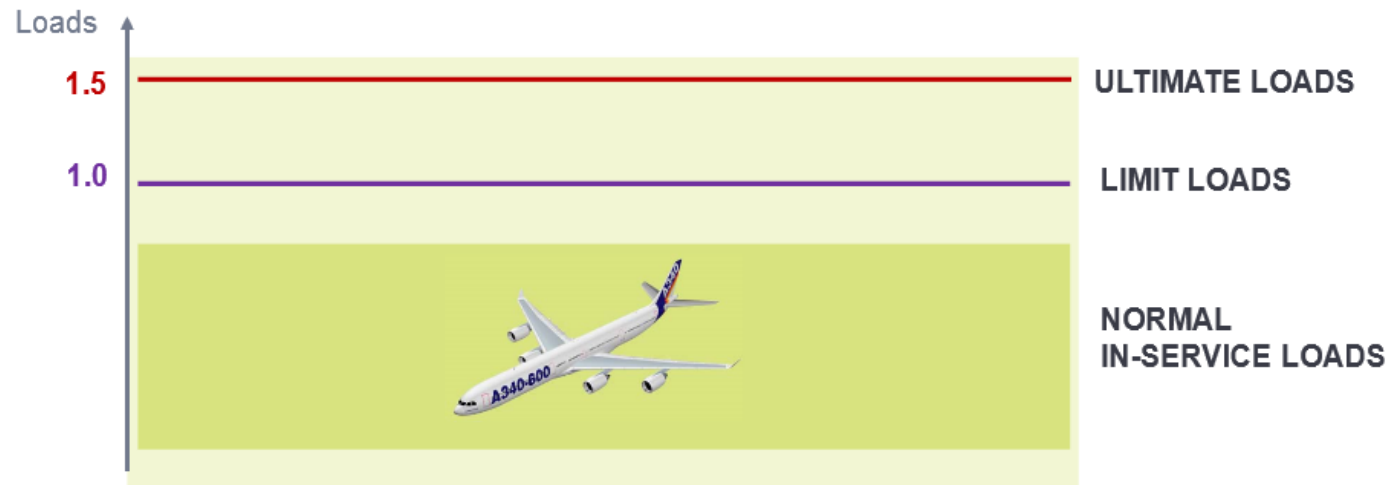
Structural Integrity

CS/FAR25.303 Factor of Safety

SF covers uncertainties and variability of loads, production, analysis methods, material characteristics, degradation during service life

Limit loads are the maximum expected load during the lifetime of the aircraft with a probability of occurrence of $1E-5$ FH/FC. The safety factor decreases the occurrence probability up to $1E-8$ or $1E-9$ FH/FC.

Structural Integrity



Structural Integrity

There are several ways to get the required structural integrity of a discrete joint:

-Fail safe joint assembly:

The joint is an assembly of more than one fitting, where under the case of failure of one of the fitting the remaining fittings are able to withstand the limit loads without failure. Without failure the assembly shall withstand the Ultimate Loads.

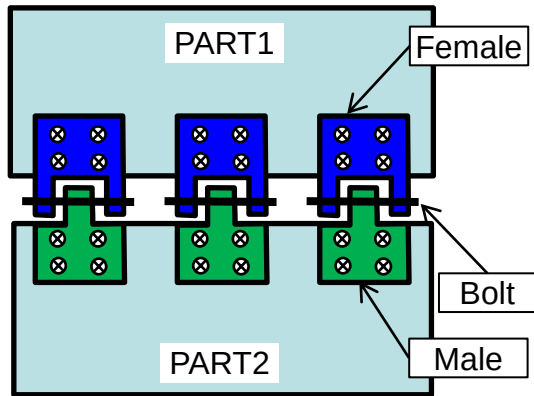
-Fitting and bolt fail-safe:

The fitting is a double fitting and the bolt is also double, bolt plus bushing, under a single failure of one of the semi fittings or one of the bolts, the fitting keeps complying with its function and is able to withstand the limit loads. Without failure the assembly shall withstand the Ultimate Loads.

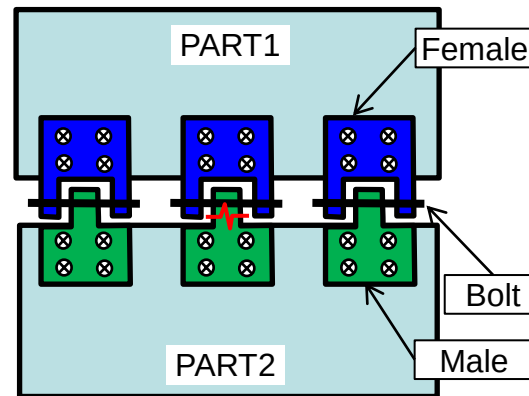
Structural Integrity

-Fitting fail safe and bolt safe-life:

The fitting is a double fitting, under the failure of a semi-fitting the other half keeps complying with its functionality. The bolt fatigue life is calculated and it must be replaced before it reaches the fatigue life. Without failure the assembly shall withstand the Ultimate Loads.

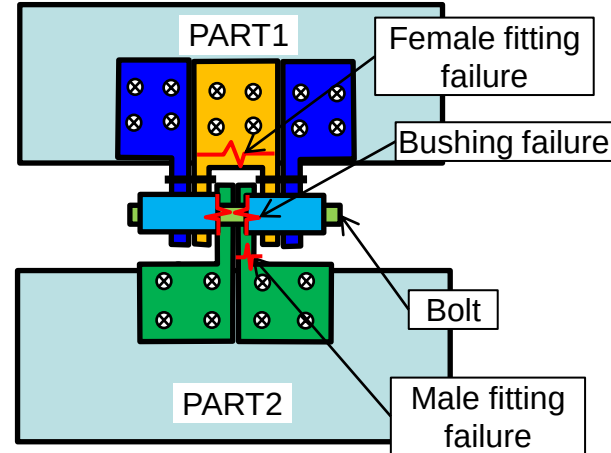
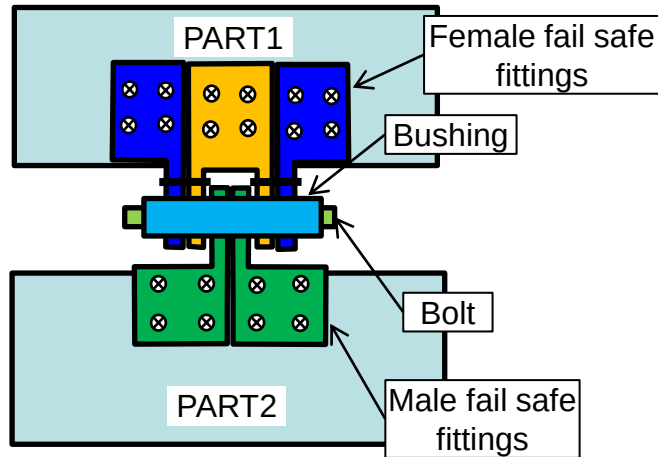
Structural Integrity: Fail safe assembly**Intact condition:**

The assembly withstands the Ultimate Loads without failure of any component

**Failure condition:**

Under failure of one fitting the remaining fittings are able to withstand the Limit Loads without failure.

Structural Integrity: Fail safe fittings and bolt



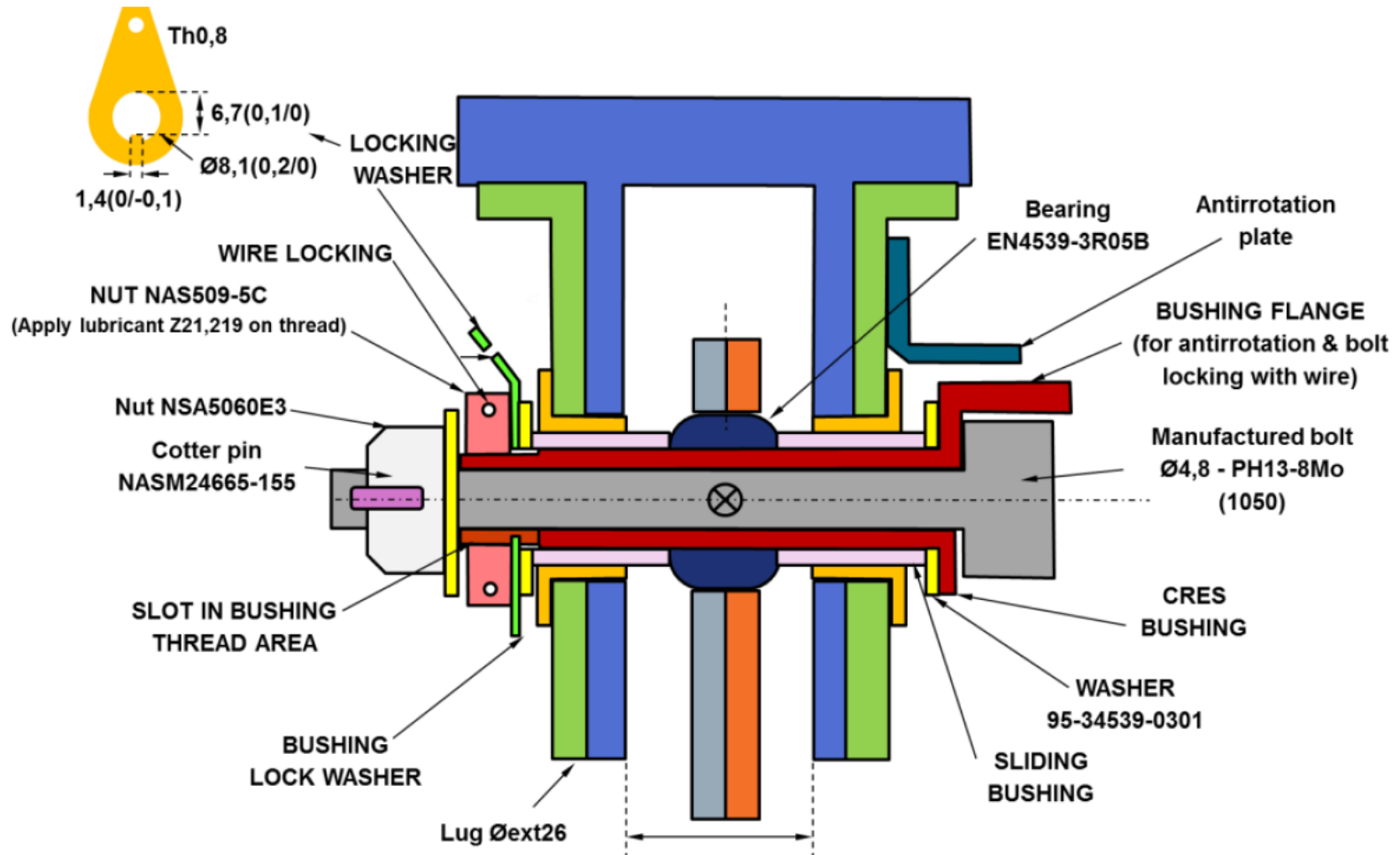
Intact condition:

The assembly withstands the Ultimate Loads without failure of any component

Failure condition:

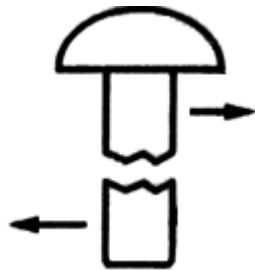
Under the failure of one of the semi-fittings or one of the bolts, the remaining semi-fittings are able to withstand the Limit Loads without failure

Structural Integrity: Fail safe fitting and bolt – Joint Detail

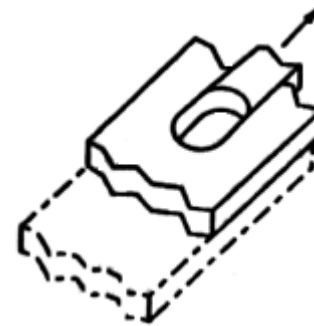


Failure Modes

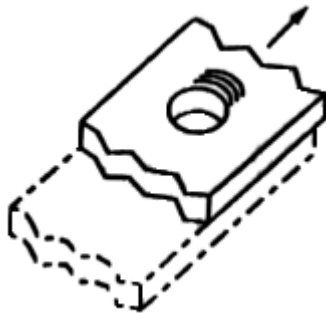
Fastener Shear: Shear failure through the fastener shank



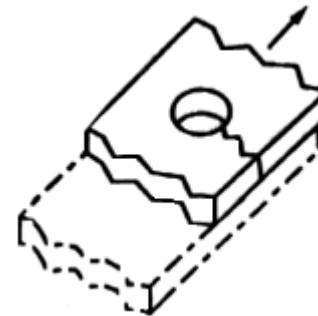
Shear-out: Shear failure of structure



Bearing: Local yield of structure



Tension: Tension failure of structure



Fastener Shear

For thin sheets the knife edge effect reduce the shear allowable

Countersunk fasteners: Shear allowable values from Tests

Two shear configurations:

$$P_{su} = F_{su} \cdot \pi \cdot D^2 / 4$$



SINGLE SHEAR

- NOT SYMMETRICAL, ADDITIONAL OVER TURNING MOMENT
- SINGLE SHEAR FACE, LESS EFFICIENT USE OF FASTENER

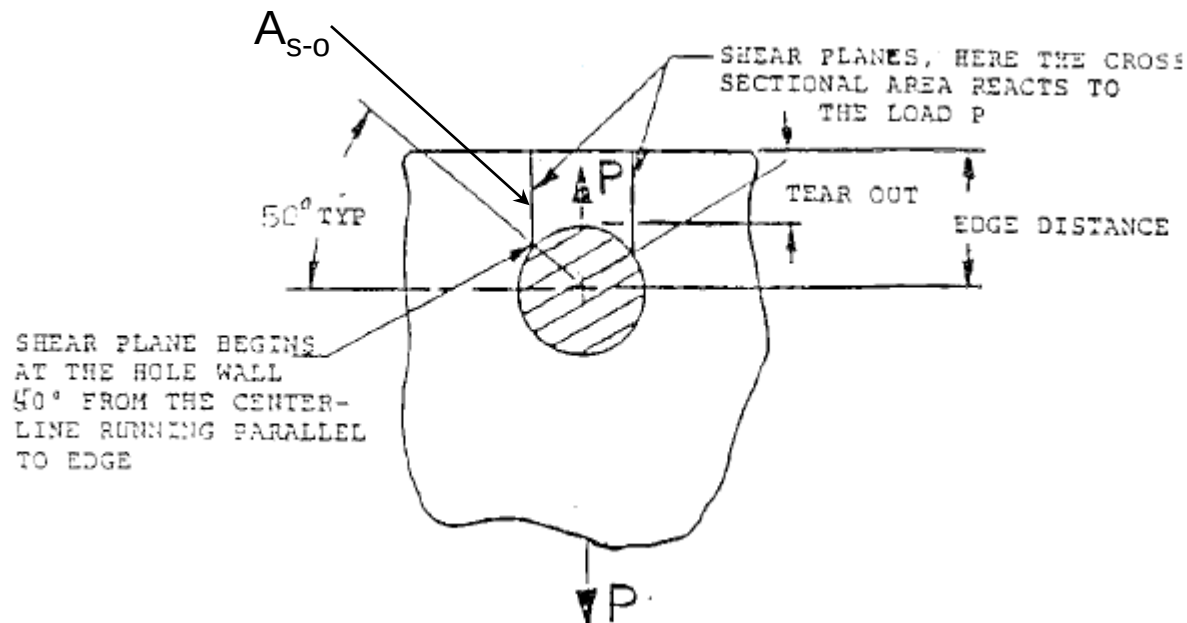


DOUBLE SHEAR

- REDUCES OVERTURNING MOMENT DUE TO SYMMETRY
- TWO SHEAR FACE S, MORE EFFICIENT USE OF FASTENER

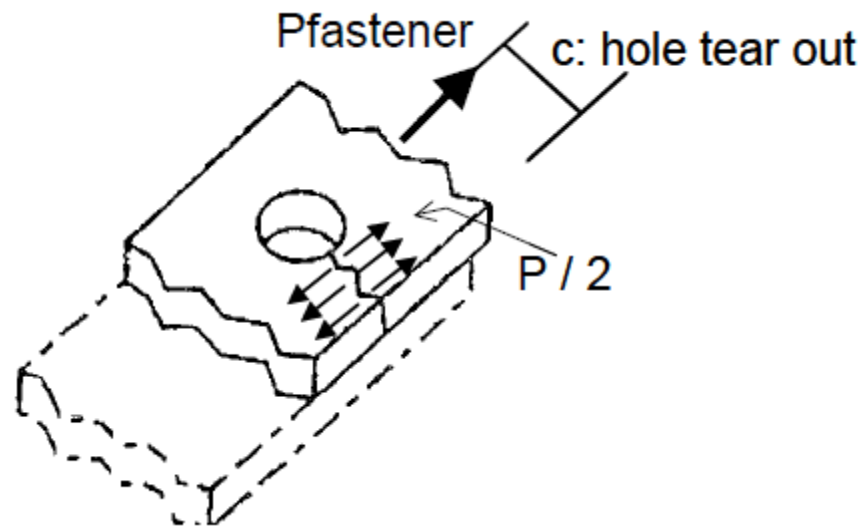
$$P_{su} = 2 \cdot F_{su} \cdot \pi \cdot D^2 / 4$$

Shear-out



$$P_{S-O} = 2 \cdot A_{S-O} \cdot t \cdot F_{su}$$

Tension failure



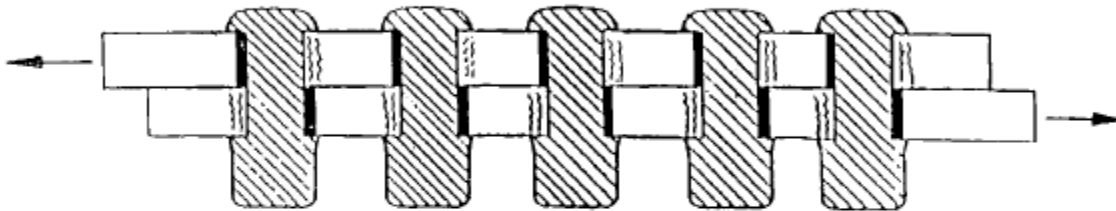
$$P_t = 2 \cdot C \cdot t \cdot F_{tu}$$

$$C = e - D/2$$

Fastener Shear Vs Plate bearing

BEARING CRITICAL:

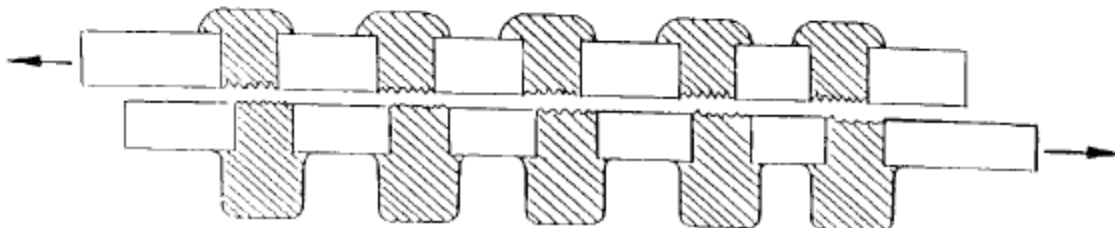
- HOLES WILL YIELD (ELONGATE) BEFORE FASTENERS FAIL IN SHEAR
- ALLOWS ALL FASTENERS TO BEAR UP (CARRY LOAD).
- FAILURE MODE IS GRADUAL



$$P_{bru} = F_{bru} \cdot D \cdot t$$

SHEAR CRITICAL:

- FASTENERS FAIL IN SHEAR PRIOR TO HOLE YIELDING
- SPlice WILL "UNZIP"
- FAILURE MODE IS SUDDEN

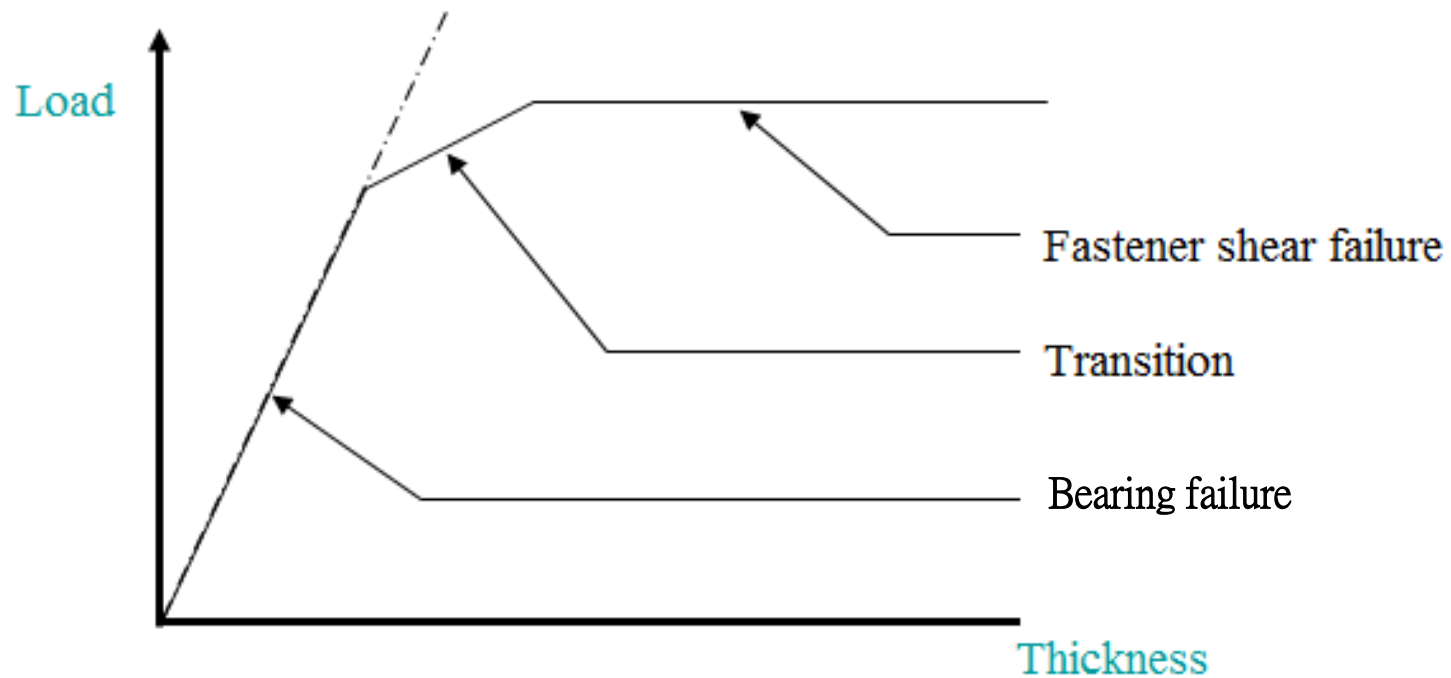


$$P_{su} = F_{su} \cdot \pi \cdot D^2 / 4$$

Typical Failure modes

Provided the distances design criteria (edge distance and fasteners pitch) are taken into account in the joint design, the joint failure will be the bolt shear, the plate bearing or a combination of both with the previously mentioned turning moment which leads to fastener head tension

Typical Failure modes

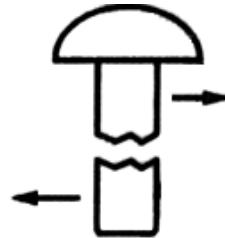


General rules:

- Close tolerance fit is recommended but not interference fit
- Expansive shank fasteners are forbidden
- Solid rivets restricted use
- On CFRP preferred fasteners material is Titanium but CRES and Monel are acceptable
- On CFRP aluminium, Inconel or Cadmium fasteners are forbidden
- Fasteners must be wet installed

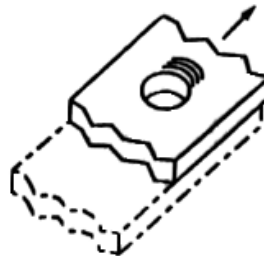
Failure modes:

Fastener shear:



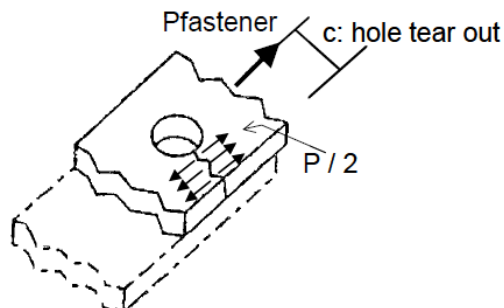
$$P_{su} = F_{su} \cdot \pi \cdot D^2 / 4$$

Bearing:



$$P_{bru} = F_{bru} \cdot D \cdot t$$

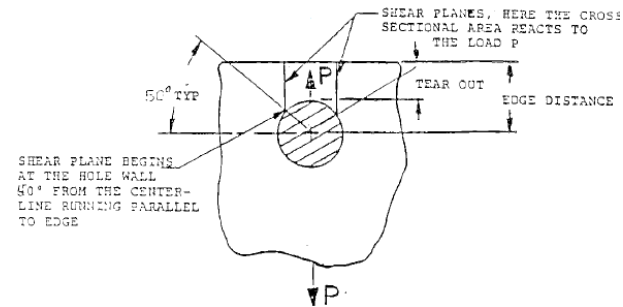
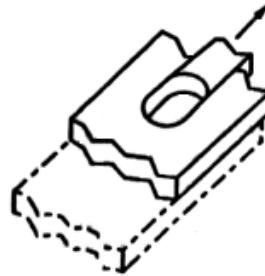
Tension:



$$P_t = (\text{Pitch} - D) \cdot t \cdot F_{tu}$$

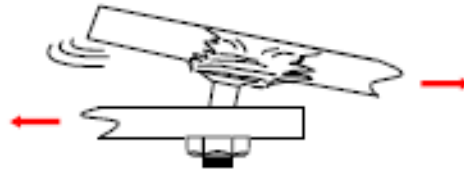
Failure modes:

Laminate shear-out:



$$P_s = 2 \cdot A_s \cdot t \cdot F_{su}$$

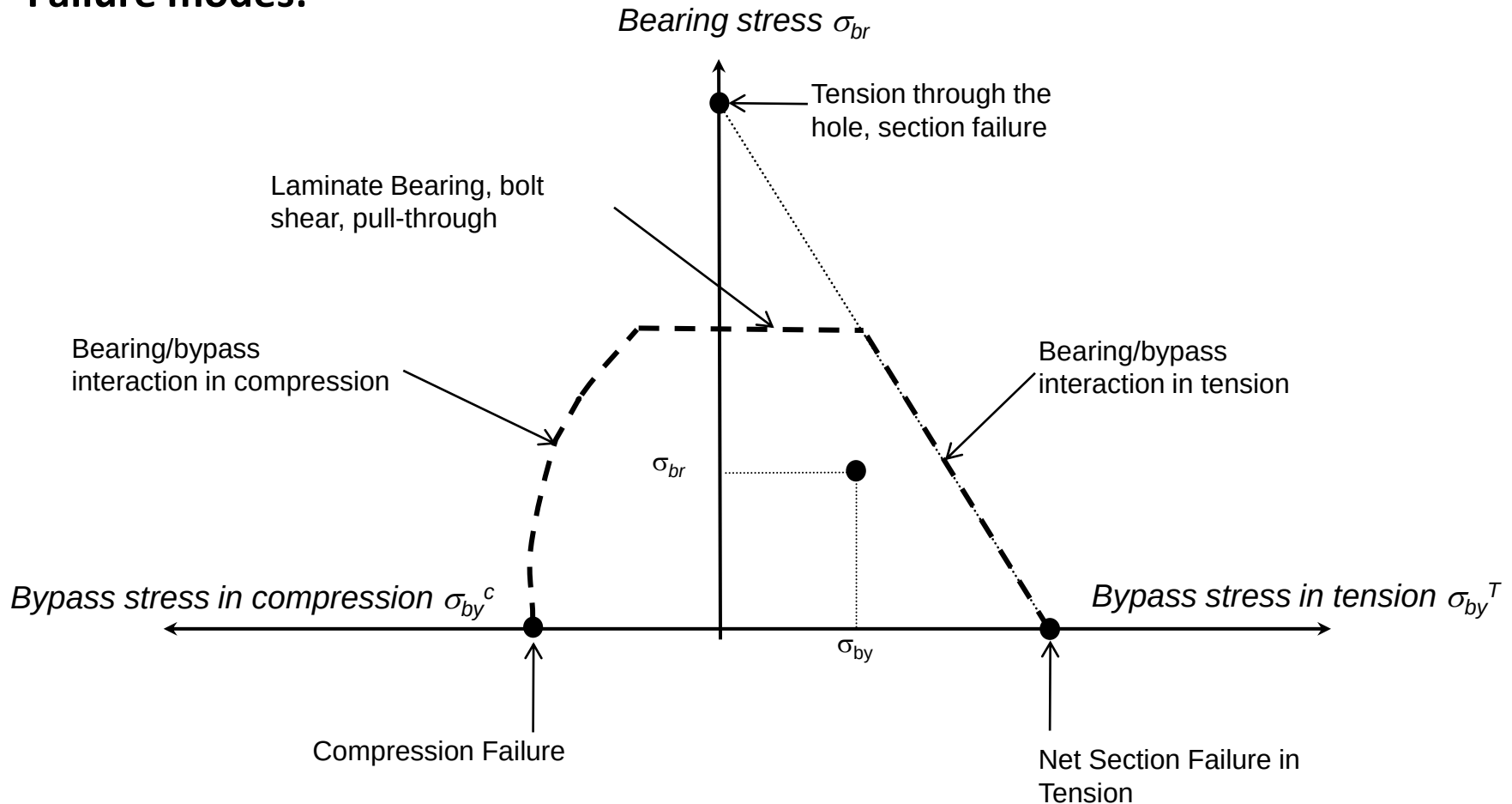
Pull-out:



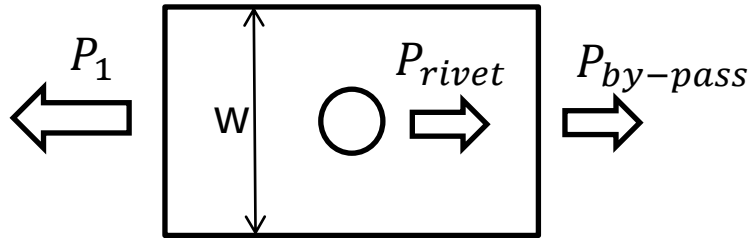
$$P_{\text{pull-out}} = ILSS \cdot \pi \cdot D_{\text{bolt-head}} \cdot t$$

ILSS: Inter Laminar Shear Strength

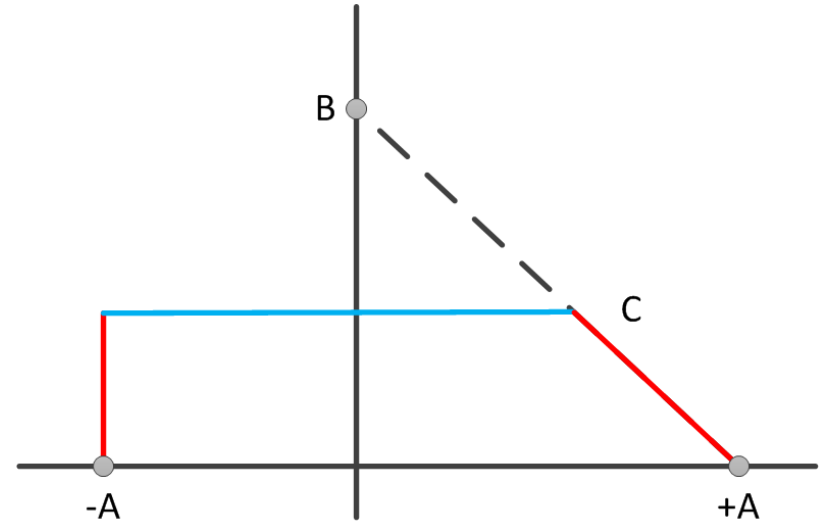
Failure modes:



Failure modes:



$$P_1 = P_{rivet} + P_{by-pass}$$



By-pass microstrains:

$$\varepsilon_{by-pass} = \frac{P_{by-pass}}{E \cdot w \cdot t}$$

Bearing stress:

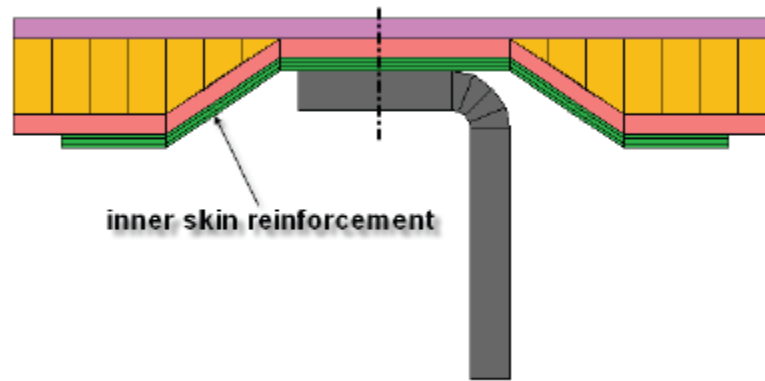
$$\sigma_{bearing} = \frac{P_{rivet}}{D \cdot t}$$

RF = MIN

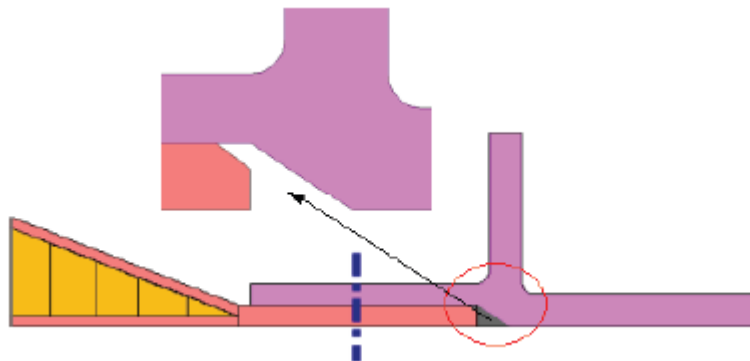
$$\left. \begin{aligned} RF_{A-B} &= \frac{1}{\left(\frac{\varepsilon_{by-pass}}{A \cdot MKDF} \right) + \left(\frac{\sigma_{bearing}}{B \cdot MKDF} \right)} \\ RF_C &= \frac{C \cdot MKDF}{\sigma_{bearing}} \end{aligned} \right\}$$

Sandwich structure joining solutions:

A) Monolithic area inside a sandwich panel for joining: Used for high load transfer, local reinforcement layers can be added.



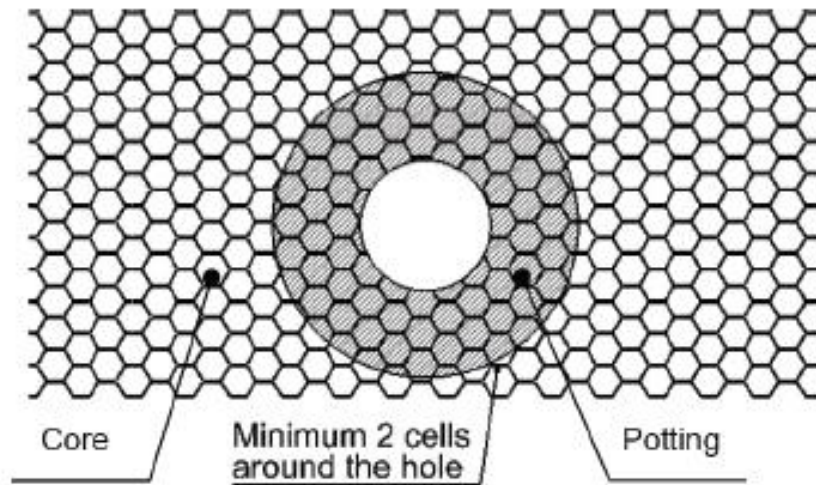
B) Monolithic area at the edge of a sandwich panel for joining:



Sandwich structure joining solutions:

C) Insert installation in sandwich core:

1°) Potting in core during manufacturing process. For high density cores the potting could be avoided

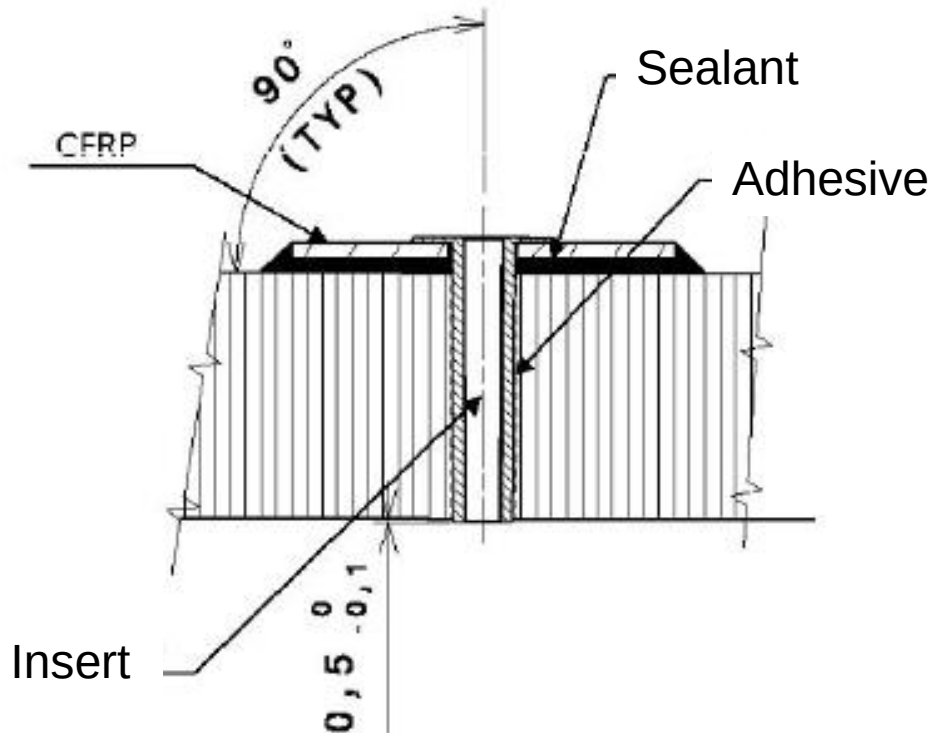


2°) Hole drilling at finished part

Sandwich structure joining solutions:

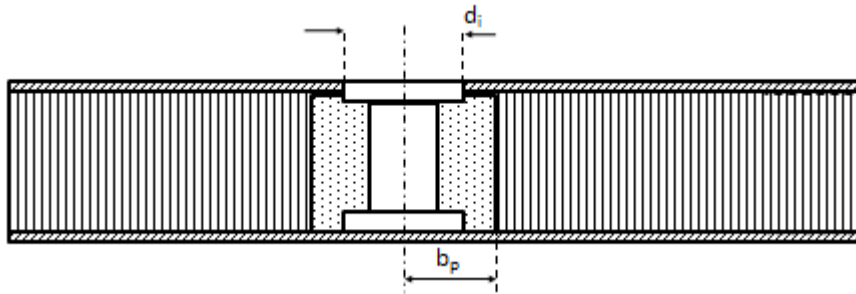
C) Insert installation in sandwich core:

3°) Insert installation

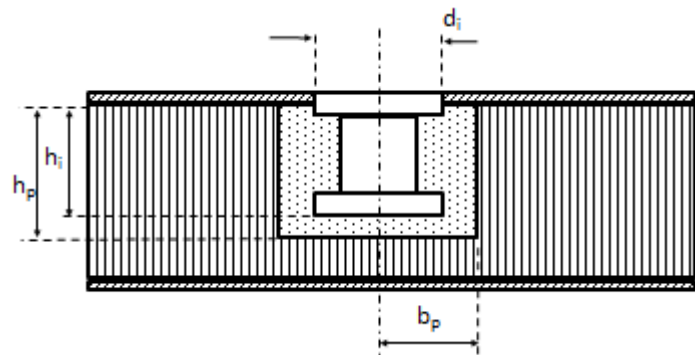


Sandwich structure joining solutions:

C) Insert installation in sandwich core:



Fully Potted Inserts



Partially Potted Inserts

1. NT-0-ANC-09015 “Estandares aplicables a C295 CN235 y C212”
2. TAE-T-NT-150063 “FT4B Design HTP & VTP Structure”
3. CAN40027 “Taladros para remaches no métricos”
4. Antonio E. Jimenez Gahete, “Curso de Estructuras en Materiales Compuestos”, Feb 2012.